

Eye Movements, Academic Level and Domain Match in Scientific Reading

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Abstract

This project investigates how eye movements during scientific reading differ depending on reader’s academic level and whether the text matches their discipline. Building on the PoTeC dataset, we refine the original expert/non-expert distinction into four reading classes: Expert, Graduate out-of-domain, Undergraduate in-domain and Undergraduate out-of-domain. We compare these classes using seven interpretable eye-movement measures covering first-pass processing, total reading effort, rereading and regressions. We first compare raw eye-movement profiles across the four classes then test whether these patterns remain stable after controlling for word-level properties. We also perform statistical comparisons between classes, analyze expert technical terms separately and use logistic regression both to evaluate predictability and to inspect which measures contribute to the predictions. Our results suggest that domain match is more clearly reflected in eye-movement behavior than academic level. Expert readers generally show the lowest values across the selected measures while out-of-domain reading tends to show higher reading effort. These patterns are also visible in the analysis of expert technical terms. Prediction results further support this conclusion: domain match is predicted better than academic level or the full four-class label.¹

Introduction

Eye movements during reading give detailed information about how readers process written language. Measures such as fixations, rereading, and regressions are related to the moment-by-moment processing of text during reading (Rayner, 1998). This makes eye-tracking useful for studying scientific reading because understanding a scientific text may depend not only on reading ability but also on familiarity with the topic and the technical terms used.

This project is based on EyeBench (Shubi et al., 2025), a benchmark that uses eye-tracking data to decode cognitive and linguistic information from reading behavior. We focus on the domain expertise task which is based on the Potsdam Textbook Corpus “PoTeC” (Jakobi et al., 2025). PoTeC is a German eye-tracking corpus where biology and physics students read scientific texts from both domains. The dataset includes information about the discipline of the reader, their level of study, the domain of the text and more informations. The original PoTeC paper uses this design to compare expert and non-expert scientific reading. In PoTeC, expert reading is defined as the case where a graduate student reads a text from his own discipline. All other cases are considered as non-expert reading. Using linear mixed models they found that expert reading was linked to shorter reading times, especially total fixation time and rereading time, as well as fe-

wer regressions. They also found that longer words, lower-frequency words, words with higher surprisal and expert technical terms generally required more processing effort. In addition their analysis of comprehension and background questions showed that expert readers usually answered more accurately than most non-expert readers. This suggests that domain expertise helps both reading behavior and understanding of scientific texts.

These results show that the expert/non-expert distinction is meaningful. However this distinction can also hide differences inside the non-expert group. In PoTeC, non-expert includes several different cases. For example a graduate student reading outside their own domain is not an expert for that text but still has advanced academic experience and may be more familiar with scientific texts. In contrast an undergraduate student reading a text from their own domain may not have advanced expertise but may still know some specific terminology related to his domain. These two situations are both classified as non-expert but they may not lead to the same reading behavior.

This leads to our main research question: **How do eye movements differ between readers depending on their academic level and whether the text matches their discipline?**

To study this question we split the original expert/non-expert division into four reading classes: expert readers, graduate out-of-domain readers, undergraduate in-domain readers and undergraduate out-of-domain readers. This allows us to compare how eye-movement patterns change with full domain expertise, academic level and domain match.

We compare the four reading classes using interpretable eye-movement measures that capture different aspects of reading behavior including first pass processing, total reading effort, rereading and regressions. Since reading times are also influenced by the properties of the words we include a control analysis for word length, lemma frequency and surprisal. We also examine expert technical terms separately since they are central to scientific reading. Finally we use prediction to test whether the selected eye-movement measures are more informative about domain match, academic level or the full four-class distinction.

The goal of this project is not simply to reproduce the original EyeBench domain expertise classification task or the original PoTeC expert/non-expert analysis. Instead we use the structure of PoTeC to refine the expert/non-expert contrast and to better understand how eye movements during scientific reading reflect full domain expertise, academic level and domain familiarity. Our results suggest that domain match plays an important role, while academic level is less clearly reflected in the eye-movement measures we selected.

Data

We use the PoTeC dataset (Jakobi et al., 2025) which is publicly available at <https://github.com/DiLi-Lab/PoTeC>. The dataset contains eye-tracking recordings from 75 native German speakers studying either biology or physics at the University of Potsdam. Participants belong to one of four groups: 16 undergraduate biology students, 27 graduate biology students, 12 undergraduate physics students and 20 graduate physics students. Undergraduate participants are first-semester BSc students while graduate participants are enrolled in MSc or PhD programs.

The reading materials consist of 12 short scientific textbook passages, six from biology and six from physics. The texts were taken from German university textbooks and are approximately 150 words long. Each participant read texts from both domains. This makes the dataset suitable for comparing in-domain and out-of-domain scientific reading.

PoTeC follows a $2 \times 2 \times 2$ fully crossed design. The three factors are reader discipline, academic level and text domain. Reader discipline and academic level vary between participants while text domain varies within participants because every reader read both biology and physics texts.

The dataset provides eye-movement measures where each word is treated as an area of interest. These measures include indicators of first-pass reading, total effort, rereading and regressions. For example first fixation duration, first-pass reading time, total fixation time, total fixation count, rereading time, regression path duration or regression counts. These measures make possible to study different aspects of reading behavior.

Technically, the dataset is organized at the word level. Each row corresponds to one word read by one participant in one text. This means that the same word-level measures are repeated across many reader-text observations and that later analyses can be aggregated at different levels, for example by reading class, reader-text pair or simply by reader.

In addition PoTeC includes many word- and text-level linguistic attributes. From lexical properties such as word length, lemma frequency and part-of-speech information to several surprisal estimates. Surprisal is defined as the negative log probability of a word given its context and in PoTeC it was estimated using several German LLMs. The dataset also includes manual annotations indicating whether a word is an expert technical term or a general technical term. Finally the dataset includes participant and trial metadata such as reader ID, text ID, reader discipline, academic level, text domain, age or gender.

Experiments and Results

This section presents our analysis of how eye movements differ across the four reading classes. We first describe how the reading classes and eye-movement measures were defined. Then, we compare raw eye-movement profiles, test whether these profiles remain stable after controlling for word properties, and perform statistical comparisons between classes.

We also analyze expert technical terms separately because they are especially relevant for our case. Finally, we use pre-

diction as a complementary analysis to determine whether eye-movement measures predict domain match, academic level, or the full four-class distinction more clearly. We also examine differences between the measures to better understand which aspects of eye movement are most informative for these predictions.

Analysis Setup

The first step of our analysis is to reorganize the original PoTeC expert/non-expert label. In the original paper, expert reading corresponds to graduate readers reading texts from their discipline while all other cases are non-expert. This definition is useful but it does not make enough distinctions for our research question.

Then we define four reading classes, shown in *Table 1*. These classes combine two factors: academic level and domain match. Academic level separates graduate and undergraduate readers while domain match indicates whether the text domain matches the reader’s discipline. This allows us to compare expert reading not only with a general non-expert group but with more specific reader-text conditions.

It is important to note that these classes are not four independent groups of participants. They are reading conditions. For example, a graduate biology reader belongs to the expert condition when reading biology texts and to the graduate out-of-domain condition when reading physics texts.

For the eye-movement analysis we select seven word-level measures that cover several standard families of measures used in reading research. Previous work (Clifton Jr et al., 2007) describes common measures such as first fixation duration, first-pass reading time, regression-path duration, regressions out and total reading time. They also show that different linguistic effects can appear in different parts of the eye-movement record, which motivates using several measures rather than relying on only one reading-time measure. This choice also follows the logic of the original PoTeC analysis which examined first-pass reading time, total fixation time, rereading time and regressions when comparing expert and non-expert reading.

We divide the selected eye-movement measures into four families. The first family captures first-pass processing. This includes FFD, the duration of the first fixation on a word during first-pass reading, and FPRT, the total time spent on the word during first-pass reading.

The second family captures total reading effort. This includes TFT, the total fixation time on the word across the whole trial and TFC, the total number of fixations on the word.

The third family captures rereading. This includes RRT, the time spent rereading the word after first-pass reading.

The fourth family captures regressions. This includes RPD_exc, the time spent in regressions after leaving the word without counting fixations on the word and TRC_out, the number of regressions initiated from the word.

We choose TRC_out rather than TRC_in because regressions launched from a word are easier to interpret as a sign of difficulty at that point when regressions into a word may be caused by difficulty that appears later in the sentence. We also choose RPD_exc rather than RPD_inc because RPD_inc includes time spent fixating the word itself.

Reading class	Academic level	Domain relation	Definition
Expert	Graduate	In-domain	Graduate reader reading a text from their own discipline
Graduate out-of-domain	Graduate	Out-of-domain	Graduate reader reading a text from the other discipline
Undergraduate in-domain	Undergraduate	In-domain	Undergraduate reader reading a text from their discipline
Undergraduate out-of-domain	Undergraduate	Out-of-domain	Undergraduate reader reading a text from other discipline

Table 1: Definition of the four reading classes. The classes are reader-text conditions and not four fully independent participant groups.

Since this information is already captured by TFT, RPD_exc provides a less redundant measure. All of these four families allow us to analyze reading behavior more completely. We focus on continuous or count measures rather than binary indicators because they provide more graded information for comparing the four reading classes.

For the descriptive analysis, we use the word-level data to show the general profile of each reading class. For the statistical tests and prediction analyses, we average the data by reader and text when needed so that the comparisons are not dominated by repeated word observations from the same reader or text.

Raw Measures Across Reading Classes

We first examine the raw eye-movement profiles of the four reading classes. The goal is to obtain a descriptive view of the data before applying statistical tests and to identify which measures show differences between classes.

To do this, we group the word-level observations by reading class and compute the mean value of each selected measure. The results are shown in Table 2. Duration measures are reported in milliseconds, while TFC and TRC_out are count measures.

Reading class	FFD (ms)	FPRT (ms)	TFT (ms)	TFC (count)	RRT (ms)	RPD_exc (ms)	TRC_out (count)
Expert	164.9	238.4	566.2	2.5	327.8	288.4	0.4
Graduate out-of-domain	170.0	259.6	696.4	3.1	436.8	389.9	0.4
Undergraduate in-domain	164.3	241.0	606.3	2.8	365.3	324.1	0.4
Undergraduate out-of-domain	165.1	247.2	664.7	3.0	417.5	371.2	0.5

Table 2: Mean raw eye-movement measures by reading class. Expert reading shows the lowest values on most measures, while out-of-domain reading generally shows higher total effort and rereading.

The clearest pattern is that the expert class has the lowest values on most measures. This is especially visible for total effort measures such as TFT and TFC and for rereading and regression measures such as RRT and RPD_exc. For example, mean TFT is 566.2 ms for expert reading compared with 696.4 ms for graduate out-of-domain reading and 664.7 ms for undergraduate out-of-domain reading.

The graduate out-of-domain class often has the highest values on total effort and rereading measures. This suggests that advanced academic level does not necessarily lead to lower reading effort when the text belongs to another disc-

ipline. In contrast, the undergraduate in-domain class is often closer to the expert condition than to the out-of-domain class.

The differences are smaller for the first-pass measures especially FFD which ranges only from 164.3 ms to 170.0 ms across the four classes. Overall, this basic analysis suggests that the main differences between reading classes appear more clearly in total effort, rereading and regression than in first-pass measures.

Control Analysis for Word Properties

Reading times are not only influenced by the reader or the text domain but also by properties of the words themselves such as word length and word-lemma frequency, which have been shown to influence eye movements (Kliegl et al., 2004). For this reason we perform a control analysis using word length, lemma frequency and surprisal. Surprisal measures how unexpected a word is in its context, usually defined as $-\log P(w_i | \text{context})$. A word with low contextual probability has high surprisal. Previous work has shown that language-model surprisal is related to human reading times (Goodkind and Bicknell, 2018).

For the surprisal variable, PoTeC provides estimates computed with different German language models. Instead of choosing only one model, we average the surprisal estimations into a single variable. This gives us one general surprisal control for each word observation.

To perform the control analysis, we use a regression. For each measure, we fit a linear regression predicting the measure from word length, lemma frequency and surprisal. We then use the residuals of this regression to compute adjusted values so this removes the part of each measure that can be explained by these word properties. The goal is to check whether the differences between reading classes remain visible after controlling for these properties.

The full adjusted class-level means are reported in Appendix Table A1. They are very close to the raw means presented in Table 2. To summarize the size of the adjustment, Figure 1 shows the average relative change between raw class-level means and the adjusted class-level means for each measure. The effect of the controls is very small for all measures. Even the largest change observed for RPD_exc remains below 1%. For most measures, the change is around 0.2% or less. This means that the class-level profiles are almost unchanged after the adjustment.

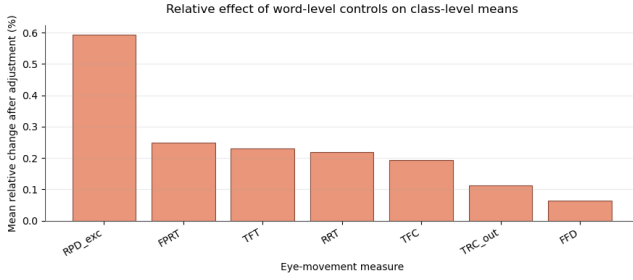


Figure 1: Relative effect of word-level controls on class-level means. The bars show the average relative change between raw and adjusted class-level means after controlling for word length, lemma frequency, and mean text surprisal. All changes are below 1%, showing that the class-level patterns remain stable after adjustment.

Overall, this control analysis shows that the patterns observed in the raw analysis are stable. The raw differences between reading classes do not seem to be driven by simple differences in word length, lemma frequency or surprisal.

Statistical Comparisons Between Reading Classes

We now test whether the differences between classes were statistically reliable. We use the raw measures because we just saw that the adjusted values were almost identical to the raw ones.

Before running the tests we average the data so each word is not counted as an independent observation. This is important because many words were read by the same reader and belonged to the same text so the word-level rows were not fully independent. For the global tests we use reader-text averages. For pairwise comparisons we use reader-class averages so each reader contributed one value per class.

We first ran Kruskal–Wallis tests for each measure to check whether there was an overall difference between the four classes. Results are shown in *Table 3*. All measures were significant except FFD, all other measures have p -values below 0.05 while FFD was close to significance but did not reach the threshold ($p = 0.059$).

Measure	Kruskal H	p -value
FFD	7.46	0.059
FPRT	17.37	< 0.001
TFT	42.81	< 0.001
TFC	41.38	< 0.001
RRT	37.12	< 0.001
RPD_exc	34.06	< 0.001
TRC_out	13.41	0.004

Table 3: Kruskal–Wallis global tests across the four reading classes using all words. Most measures show significant class differences.

We then run the six pairwise comparisons between the four classes. We use Wilcoxon tests for paired comparisons involving the same readers such as Expert vs. Graduate out-of-domain and Undergraduate in-domain vs. Undergraduate out-of-domain. For the other comparisons we use Mann–Whitney U tests. Expert vs. Graduate out-of-domain test

results are shown in *Table 4*. Other comparison results are reported in *Appendix Table A2*.

Measure	Test	Mean A	Mean B	Difference A–B	p -value
FFD	Wilcoxon signed-rank	164.87	170.05	−5.18	< 0.001
FPRT	Wilcoxon signed-rank	238.32	259.54	−21.22	< 0.001
TFT	Wilcoxon signed-rank	566.52	695.90	−129.38	< 0.001
TFC	Wilcoxon signed-rank	2.54	3.06	−0.52	< 0.001
RRT	Wilcoxon signed-rank	328.19	436.36	−108.17	< 0.001
RPD_exc	Wilcoxon signed-rank	288.76	389.48	−100.72	< 0.001
TRC_out	Wilcoxon signed-rank	0.38	0.44	−0.06	0.006

Table 4: Pairwise comparison between Expert and Graduate out-of-domain reading using all words with the Wilcoxon signed-rank test. Expert reading is significantly lower on all measures showing a clear domain-match effect for graduate readers.

The clearest difference is in the Expert vs. Graduate out-of-domain comparison. This comparison tests the same graduate readers inside and outside their discipline. Expert reading is significantly lower on all seven measures, with $p < 0.001$ for all measures except TRC_out which is still statistically significant ($p = 0.006$). This shows that graduate readers read with much less effort when the text matches their discipline.

The comparisons involving Expert vs. Undergraduate in-domain and Graduate out-of-domain vs. Undergraduate out-of-domain suggest that academic level does not explain the reading profiles. Expert reading is not significantly different from Undergraduate in-domain reading, for example $p = 0.283$ for TFC or $p = 0.747$ for FFD. Similarly Graduate out-of-domain is not significantly different from Undergraduate out-of-domain on any measure with all p -values greater than 0.3. This suggests that undergraduate readers reading inside their domain are close to experts while graduate readers reading outside their domain are close to undergraduate out-of-domain readers.

The remaining comparisons support the same idea. Expert readers show significantly lower values than Undergraduate out-of-domain readers for TFT ($p = 0.040$), TFC ($p = 0.021$), RRT ($p = 0.042$), and RPD_exc ($p = 0.043$). Undergraduate in-domain readers also show significantly lower values than Undergraduate out-of-domain readers for TFT ($p = 0.020$), TFC ($p = 0.032$), RRT ($p = 0.022$), and RPD_exc ($p = 0.019$) while TRC_out is close to significance ($p = 0.056$). In contrast, Graduate out-of-domain and Undergraduate in-domain are not significantly different with all p -values greater than 0.140, even though the Graduate out-of-domain class has higher values descriptively.

Overall, the results suggest that domain match matters more clearly than academic level. This pattern is especially visible in measures related to total effort, rereading and regression while first-pass measures show weaker and less consistent differences. Readers who are in-domain, even undergraduate readers, are closer to experts than graduate readers outside their domain.

Expert Technical Words Analysis

We also want to analyze the words annotated in PoTeC as expert technical terms. The goal is to check whether the same

class patterns appear on the words that are most specific to the scientific domain.

We filter the dataset to keep only expert technical words. We first compute the mean values for each class reported in Table 5. We then repeat the same statistical procedure that we did before: Kruskal–Wallis tests for overall differences between the four classes shown in Table 6, followed by pairwise comparisons between all classes. Expert vs. Graduate out-of-domain test results are shown in Table 7. Other comparison results are reported in Appendix Table A3.

Reading class	FFD (ms)	FPRT (ms)	TFT (ms)	TFC (count)	RRT (ms)	RPD_exc (ms)	TRC_out (count)
Expert	209.3	508.2	1185.0	5.1	676.9	428.2	0.6
Graduate out-of-domain	215.1	645.9	1699.7	7.0	1053.8	518.4	0.7
Undergraduate in-domain	208.5	564.3	1414.0	6.2	849.7	344.3	0.7
Undergraduate out-of-domain	210.7	624.5	1622.3	6.8	997.8	640.0	0.7

Table 5: Mean eye-movement measures on expert technical words by reading class. Expert technical words lead to much higher reading effort than average words, especially for out-of-domain readers.

The measures means show that expert technical words are as expected, much harder to process than the average word. For example, the mean TFT for expert readers on all words is 566.2 ms, but it increases to 1185.0 ms on expert technical words.

Measure	Kruskal H	p-value
FFD	0.71	0.871
FPRT	11.92	0.008
TFT	15.70	0.001
TFC	16.82	< 0.001
RRT	13.47	0.004
RPD_exc	2.85	0.415
TRC_out	1.39	0.707

Table 6: Kruskal–Wallis global tests across the four reading classes using only expert technical words. Differences mainly appear for first-pass time, total effort and rereading but not for regression measures.

The Kruskal–Wallis tests show significant differences for FPRT, TFT, TFC and RRT. These measures capture first-pass processing, total reading effort and rereading while FFD, RPD_exc, and TRC_out are not significant in the global tests, with p -values greater than 0.4. This suggests that the strongest differences between classes appear in the time spent reading and rereading the words rather than in the very first fixation or in regressions.

Measure	Test	Mean A	Mean B	Difference A–B	p-value
FFD	Wilcoxon signed-rank	209.12	214.66	−5.54	0.038
FPRT	Wilcoxon signed-rank	502.34	631.08	−128.74	< 0.001
TFT	Wilcoxon signed-rank	1190.96	1653.00	−462.05	< 0.001
TFC	Wilcoxon signed-rank	5.12	6.81	−1.69	< 0.001
RRT	Wilcoxon signed-rank	688.62	1021.92	−333.30	< 0.001
RPD_exc	Wilcoxon signed-rank	467.08	556.98	−89.90	0.611
TRC_out	Wilcoxon signed-rank	0.63	0.70	−0.07	0.169

Table 7: Pairwise comparison between Expert and Graduate out-of-domain reading using only expert technical words with the Wilcoxon signed-rank test. Expert readers show lower reading time and rereading on technical terms, while regression measures are not significant.

The clearest pairwise result is again Expert vs. Graduate out-of-domain. Expert readers have significantly lower values for FFD ($p = 0.038$), FPRT, TFT, TFC and RRT (all have $p < 0.001$). RPD_exc and TRC_out are not significant in this comparison (p greater than 0.16). A possible explanation is that technical terms increase the time readers spend first-pass processing or rereading the word without causing them to move back to earlier words of the sentence.

Similarly to the all-words analysis, the comparison between Expert and Undergraduate in-domain is not significant on any measure. Also, Expert vs. Undergraduate out-of-domain shows clear differences for reading-effort measures: FPRT ($p = 0.034$), TFT ($p = 0.003$), TFC ($p = 0.001$) and RRT ($p = 0.005$).

The other comparisons are weaker. Graduate out-of-domain is not significantly different from Undergraduate in-domain or Undergraduate out-of-domain. Undergraduate in-domain vs. Undergraduate out-of-domain shows only one significant difference for RRT ($p = 0.040$) with in-domain readers rereading less. This points in the same direction as before but is weaker than for the graduate in-domain/out-of-domain comparison.

Overall, this analysis supports the main conclusion from the previous one. Domain match seems to matter more than academic level. Expert readers show much lower reading effort on technical terms when the text matches their discipline while graduate readers outside their domain do not clearly differ from undergraduate readers. The effect is especially visible in first-pass measures, total effort and rereading time. Regression measures are less consistent here, probably because difficulty with technical terms is mostly expressed as longer fixation or rereading of the term itself rather than as systematic movement back to earlier text.

Prediction Analysis

As a final analysis, we test whether our seven measures have enough information to predict different aspects of the reading condition. The goal is to check which distinction is more visible in the eye-movement data: domain match, academic level or the full four-class label.

For this analysis, we average the data at the reader-text level. Each observation corresponds to one reader reading one text. We compare logistic regression with a majority-class baseline. We choose to use logistic regression because it gives both prediction scores and interpretable coefficients. This allows us to evaluate whether the measures are predictive and also to see how much each measure contributes to the prediction. Evaluation is done using five-fold GroupKFold cross-validation grouped by reader ID, so that the same reader cannot appear in both training and test folds. This avoids testing the model on readers that it has already seen during training.

Task	Model	Balanced accuracy	Macro-F1
Domain match	Baseline	0.50	0.33
Domain match	Logistic regression	0.61	0.61
Academic level	Baseline	0.50	0.38
Academic level	Logistic regression	0.49	0.48
Four-class reading class	Baseline	0.25	0.12
Four-class reading class	Logistic regression	0.30	0.30

Table 8: Prediction results using the seven eye-movement measures. Scores are averaged across five GroupKFold splits grouped by reader ID. Domain match is predicted better than academic level or the full four-class label.

We report balanced accuracy and macro-F1. Balanced accuracy is useful here because some targets are not equally represented in the data and it averages recall across classes instead of being dominated by the majority class. Macro-F1 is also useful because it combines precision and recall and gives equal weight to each class which makes it more informative when the classes are not equally frequent. The results are shown in *Table 8*.

The results show that domain match is the most predictable target. Logistic regression gets a balanced accuracy of 0.61 and a macro-F1 of 0.61, compared with 0.50 and 0.33 for the baseline. This suggests that the measures contain a meaningful signal about whether readers are reading inside or outside their discipline.

In contrast, academic level is not predicted well. Logistic regression reaches only 0.49 balanced accuracy and 0.48 macro-F1 which is close to the baseline. This means that the selected measures do not separate graduate from undergraduate readers independently of domain match. The full four-class prediction task is also difficult: logistic regression improves over the baseline but the score remains low, with 0.30 balanced accuracy and 0.30 macro-F1.

This is expected because the previous analyses show that some reading classes are very similar, especially Expert and Undergraduate in-domain or Graduate out-of-domain and

Undergraduate out-of-domain.

To interpret the logistic regression we also inspect its coefficients shown in *Figure 2*. In the domain prediction plot positive coefficients mean that higher values of a measure push the model toward predicting out-of-domain while negative coefficients push it toward in-domain. In the academic-level prediction plot positive coefficients push the model toward predicting graduate, while negative coefficients push it toward undergraduate. The error bars show the variation across the cross-validation folds.

The clearest signal for domain match comes from first-pass processing, total effort and rereading measures. FPRT, TFT, TFC and RRT have positive coefficients meaning that they push the model toward predicting out-of-domain reading. In contrast, the regression measures are weaker and less consistent. For academic level, the coefficients are less stable across the measures and do not correspond to strong prediction scores, suggesting that academic level is not clearly captured by our measures.

Overall, this prediction analysis supports the conclusions from the previous analyses. The eye-movement measures are more informative about domain match than about academic level.

Discussion and Conclusion

In this project, we asked how eye movements differ between readers depending on their academic level and whether the text matches their discipline. We extended the original PoTeC expert/non-expert distinction by splitting the data into four classes. This allowed us to separate academic level from domain match.

Our results suggest that eye movements reflect domain match more clearly than academic level. Expert reading showed the lowest values across the selected measures while out-of-domain reading tended to show higher values, indicating greater reading effort and difficulty.

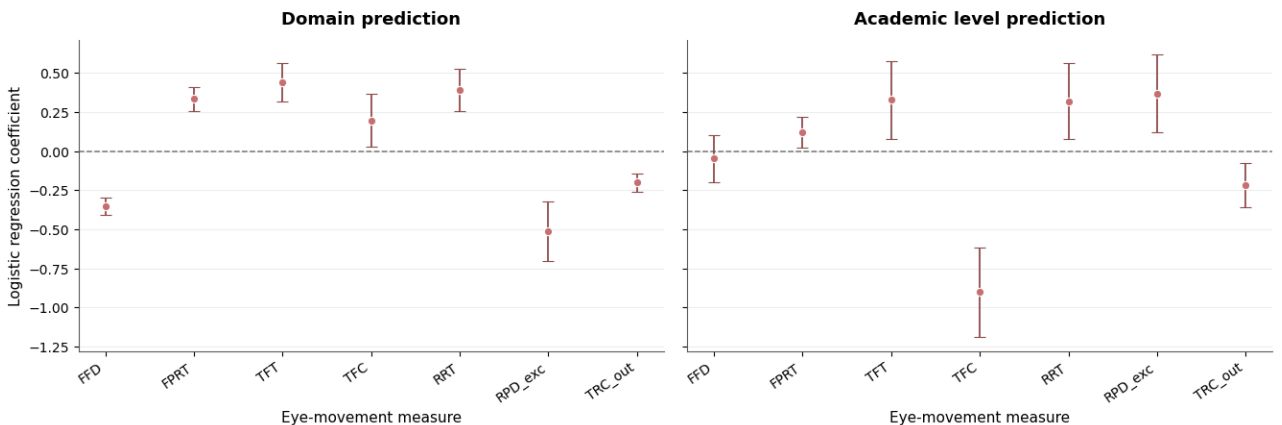


Figure 2: Logistic regression coefficients for the domain-match and academic-level prediction tasks. For the domain match model, positive coefficients indicate that higher values push the model toward predicting out-of-domain reading. For the academic level model, positive coefficients indicate that higher values push the model toward predicting graduate readers. Error bars show 95% confidence intervals across folds. The clearest signal appears for domain match.

One key pattern is that undergraduate readers reading in their domain were often closer to expert readers than to out-of-domain readers, when graduate readers reading outside their domain often showed higher reading effort.

This gives us part of the answer to our research question: **academic level may play some role but our results show a clearer effect of domain match in the eye-movement measures we analyzed.**

This interpretation was supported by several analyses that we performed. The control analysis showed that the class-level patterns remained unchanged after controlling for word length, lemma frequency and surprisal. The statistical comparison tests showed that the strongest and most consistent difference was between Expert and Graduate out-of-domain reading, where the same graduate readers read with much less effort in their domain. The expert technical-word analysis showed a similar pattern: as expected, technical terms were harder to process for all readers but expert readers showed lower effort when the terms belonged to their own domain. Finally, the prediction analysis showed that logistic regression predicted domain match better than academic level or the full four-class label, even if the prediction scores remained modest.

These conclusions should be interpreted carefully. Our analysis was based on one dataset with German academic-level texts from only two domains, biology and physics, so the results may not generalize to other languages, disciplines or text types. In addition, the four classes are reader-text conditions rather than fully independent groups and the sample size is not very large which means that some comparisons may be influenced by repeated observations from the same readers. Finally, our analysis was intentionally focused: we selected seven eye-movement measures and did not explore all possible linguistic, textual or participant-level factors.

Future work could extend this approach to larger datasets with more domains, participants or diverse texts. Another interesting direction would be to distinguish between different types of technical terms and to combine eye-movement measures with comprehension scores. It would also be useful to study the measures in more depth, for example by testing whether certain combinations of measures capture different aspects of reading expertise.

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Use of AI

We used ChatGPT (GPT-5.5) mainly to help with the code and also for a few English formulation clarifications and pre-cisions.

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Appendix

Reading class	Adj FFD (ms)	Adj FPRT (ms)	Adj TFT (ms)	Adj TFC (count)	Adj RRT (ms)	Adj RPD_exc (ms)	Adj TRC_out (count)
Expert	164.7	237.7	564.7	2.5	327.0	286.4	0.4
Graduate out-of-domain	170.2	260.3	697.9	3.1	437.6	391.9	0.4
Undergraduate in-domain	164.2	240.4	604.9	2.8	364.5	322.2	0.4
Undergraduate out-of-domain	165.2	247.8	666.1	3.0	418.3	373.2	0.5

Table A1: Adjusted mean eye-movement measures by reading class after controlling for word-level properties. Values were adjusted for word length, lemma frequency and mean text surprisal. The adjusted means are very close to the raw means showing that the main class-level patterns remain stable after controlling for these properties.

Expert vs. Undergraduate in-domain						Expert vs. Undergraduate out-of-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value	Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Mann-Whitney <i>U</i>	164.87	164.34	0.53	0.747	FFD	Mann-Whitney <i>U</i>	164.87	165.03	-0.17	0.943
FPRT	Mann-Whitney <i>U</i>	238.32	241.00	-2.67	0.978	FPRT	Mann-Whitney <i>U</i>	238.32	247.00	-8.67	0.603
TFT	Mann-Whitney <i>U</i>	566.52	606.38	-39.86	0.446	TFT	Mann-Whitney <i>U</i>	566.52	664.53	-98.01	0.040
TFC	Mann-Whitney <i>U</i>	2.54	2.78	-0.24	0.283	TFC	Mann-Whitney <i>U</i>	2.54	3.03	-0.48	0.021
RRT	Mann-Whitney <i>U</i>	328.19	365.38	-37.19	0.421	RRT	Mann-Whitney <i>U</i>	328.19	417.53	-89.34	0.042
RPD_exc	Mann-Whitney <i>U</i>	288.76	324.16	-35.39	0.402	RPD_exc	Mann-Whitney <i>U</i>	288.76	371.32	-82.55	0.043
TRC_out	Mann-Whitney <i>U</i>	0.38	0.42	-0.03	0.822	TRC_out	Mann-Whitney <i>U</i>	0.38	0.46	-0.08	0.212

Graduate out-of-domain vs. Undergraduate in-domain						Graduate out-of-domain vs. Undergraduate out-of-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value	Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Mann-Whitney <i>U</i>	170.05	164.34	5.71	0.206	FFD	Mann-Whitney <i>U</i>	170.05	165.03	5.01	0.338
FPRT	Mann-Whitney <i>U</i>	259.54	241.00	18.54	0.159	FPRT	Mann-Whitney <i>U</i>	259.54	247.00	12.54	0.355
TFT	Mann-Whitney <i>U</i>	695.90	606.38	89.52	0.144	TFT	Mann-Whitney <i>U</i>	695.90	664.53	31.37	1.000
TFC	Mann-Whitney <i>U</i>	3.06	2.78	0.28	0.180	TFC	Mann-Whitney <i>U</i>	3.06	3.03	0.04	0.943
RRT	Mann-Whitney <i>U</i>	436.36	365.38	70.98	0.210	RRT	Mann-Whitney <i>U</i>	436.36	417.53	18.83	0.900
RPD_exc	Mann-Whitney <i>U</i>	389.48	324.16	65.32	0.257	RPD_exc	Mann-Whitney <i>U</i>	389.48	371.32	18.16	0.857
TRC_out	Mann-Whitney <i>U</i>	0.44	0.42	0.02	0.372	TRC_out	Mann-Whitney <i>U</i>	0.44	0.46	-0.02	0.848

Undergraduate in-domain vs. Undergraduate out-of-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Wilcoxon signed-rank	164.34	165.03	-0.70	0.695
FPRT	Wilcoxon signed-rank	241.00	247.00	-6.00	0.465
TFT	Wilcoxon signed-rank	606.38	664.53	-58.15	0.020
TFC	Wilcoxon signed-rank	2.78	3.03	-0.25	0.032
RRT	Wilcoxon signed-rank	365.38	417.53	-52.15	0.022
RPD_exc	Wilcoxon signed-rank	324.16	371.32	-47.16	0.019
TRC_out	Wilcoxon signed-rank	0.42	0.46	-0.05	0.056

Table A2: Additional pairwise comparisons between reading classes using all words. Each block reports one comparison, with the test chosen according to the structure of the data: Wilcoxon signed-rank tests for paired comparisons involving the same readers and Mann-Whitney *U* tests for independent comparisons. These results show that Expert and Undergraduate in-domain are not significantly different while Undergraduate in-domain differs from Undergraduate out-of-domain mainly on total effort, rereading and regression measures.

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Expert vs. Undergraduate in-domain						Expert vs. Undergraduate out-of-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value	Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Mann-Whitney <i>U</i>	209.12	208.03	1.09	0.657	FFD	Mann-Whitney <i>U</i>	209.12	209.70	-0.58	0.956
FPRT	Mann-Whitney <i>U</i>	502.34	559.73	-57.39	0.226	FPRT	Mann-Whitney <i>U</i>	502.34	605.75	-103.41	0.034
TFT	Mann-Whitney <i>U</i>	1190.96	1406.17	-215.21	0.102	TFT	Mann-Whitney <i>U</i>	1190.96	1593.55	-402.59	0.003
TFC	Mann-Whitney <i>U</i>	5.12	6.10	-0.98	0.070	TFC	Mann-Whitney <i>U</i>	5.12	6.76	-1.64	0.001
RRT	Mann-Whitney <i>U</i>	688.62	846.44	-157.82	0.176	RRT	Mann-Whitney <i>U</i>	688.62	987.80	-299.18	0.005
RPD_exc	Mann-Whitney <i>U</i>	467.08	369.12	97.97	0.697	RPD_exc	Mann-Whitney <i>U</i>	467.08	686.16	-219.08	0.222
TRC_out	Mann-Whitney <i>U</i>	0.63	0.67	-0.05	0.801	TRC_out	Mann-Whitney <i>U</i>	0.63	0.75	-0.12	0.308

Graduate out-of-domain vs. Undergraduate in-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Mann-Whitney <i>U</i>	214.66	208.03	6.63	0.446
FPRT	Mann-Whitney <i>U</i>	631.08	559.73	71.35	0.116
TFT	Mann-Whitney <i>U</i>	1653.00	1406.17	246.83	0.156
TFC	Mann-Whitney <i>U</i>	6.81	6.10	0.71	0.187
RRT	Mann-Whitney <i>U</i>	1021.92	846.44	175.48	0.210
RPD_exc	Mann-Whitney <i>U</i>	556.98	369.12	187.87	0.235
TRC_out	Mann-Whitney <i>U</i>	0.70	0.67	0.03	0.665

Graduate out-of-domain vs. Undergraduate out-of-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Mann-Whitney <i>U</i>	214.66	209.70	4.97	0.611
FPRT	Mann-Whitney <i>U</i>	631.08	605.75	25.33	0.551
TFT	Mann-Whitney <i>U</i>	1653.00	1593.55	59.46	0.805
TFC	Mann-Whitney <i>U</i>	6.81	6.76	0.05	0.780
RRT	Mann-Whitney <i>U</i>	1021.92	987.80	34.12	0.706
RPD_exc	Mann-Whitney <i>U</i>	556.98	686.16	-129.18	0.595
TRC_out	Mann-Whitney <i>U</i>	0.70	0.75	-0.05	0.780

Undergraduate in-domain vs. Undergraduate out-of-domain					
Measure	Test	Mean A	Mean B	Difference A-B	p-value
FFD	Wilcoxon signed-rank	208.03	209.70	-1.67	0.937
FPRT	Wilcoxon signed-rank	559.73	605.75	-46.02	0.451
TFT	Wilcoxon signed-rank	1406.17	1593.55	-187.38	0.077
TFC	Wilcoxon signed-rank	6.10	6.76	-0.65	0.104
RRT	Wilcoxon signed-rank	846.44	987.80	-141.36	0.040
RPD_exc	Wilcoxon signed-rank	369.12	686.16	-317.04	0.099
TRC_out	Wilcoxon signed-rank	0.67	0.75	-0.08	0.227

Table A3: Additional pairwise comparisons between reading classes using only expert technical words, the table shows that most comparisons are not significant except when Expert readers are compared with Undergraduate out-of-domain readers and more weakly, when Undergraduate in-domain readers are compared with Undergraduate out-of-domain readers.